

Bump Map

Pads per EIC tile

- Tile has 4 TX slices
- Slices separated by rows of grounds
- Tile has 4 RX slices
- Also separated by rows of grounds
- Dimensions (in micron shown here)
 - 270um x 540um

	45	90	135	180	225	270
45	VDDL	MPDIN0	TXP0	VDD_PD_RX0	RXP0	DACRX0A
90	VDDH	DACTX0	TXN0	VDD_MPD_TX0	RXN0	DACRX0B
135	VDDL	SP0	GND	GND	GND	GND
180	VDDL	MPDIN1	TXP1	VDD_PD_RX1	RXP1	DACRX1A
225	VDDH	DACTX1	TXN1	VDD_MPD_TX1	RXN1	DACRX1B
270	VDDL	SP1	GND	GND	GND	GND
315	VDDL	MPDIN2	TXP2	VDD_PD_RX2	RXP2	DACRX2A
360	VDDH	DACTX2	TXN2	VDD_MPD_TX2	RXN2	DACRX2B
405	VDDL	SP2	GND	GND	GND	GND
450	VDDL	MPDIN3	TXP3	VDD_PD_RX3	RXP3	DACRX3A
495	VDDH	DACTX3	TXN3	VDD_MPD_TX3	RXN3	DACRX3B
540	VDDL	SP3	GND	GND	GND	GND

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- Questions:
 - How big are the driver and TIA?
 - What porosity (in terms of TSVs and metal KOZs) are you assuming in the PIC?
 - Hybrid bond pads do not need to be uniform
 - Are you assuming a power grid/geometry translation in the PIC to go from 45um ubump pitch to 150um C4?

Link Arch - Per Link DAC/ADC requirements

	Per Tile Component Counts
MRM PN junction DAC	4
mPDs ADC	18
VOA DAC	4
INTL Heater DAC	8
MRM heater DAC	4
CRR heater DAC	8
Pol Cntrl Heater DAC	6

*mPDs and some heaters heaters can be multiplexed to reduced the total number of ADCs/DACs

Bump Map - Photonics Constraints

Hybrid bond pads do not need to be uniform. This allows flexibility in terms of photonic placement.

Are there any KOZs on the PIC (for inductors etc)

	Size (x, y) approx)	x distance to next pad	y distance to next pad
mPD	20 x 3 um	>10um	>25 um
CRR*	75 x 55um	TBD > 100um	>65um
MRM*	55 x 35um	>=45um	>= 45um

*Assumes y is the direction of light propagation as drawn on the enclosed Bump Map. Larger spacings are likely needed to prevent thermal cross-talk

**MRM and CRR heater controls can be electrically routed to different locations on chip

Transmitter Questions

- Do we have control over MRM bias? Can this be increased?
- For the MRM bias network, can you share the R and C values, Vbias, and the TIA common mode you have in mind?
- Whats the step size granularity for each Tx tap?
- Does AthenaCore implement OCI constant-AOP squelch (modulation off, average power held, DC bias independent of swing mute), and expose squelch state and timing to an external photonic controller?

Receiver Questions

- What drives the 2 to 3 V reverse bias requirement for the high-speed PD?
- Option to have mPDs biased at nominal 0V
- How do you plan to bias the high-speed PD between RXP, RXN and VDD_PD_RX, and how will the DACs reach 2 to 3 V of reverse bias with a 1.6 V full scale?
- How do you plan to close the loop on RX ring tuning?
- What are the equalization capabilities? Is a 1-tap DFE in the plan?
- Can we go higher than 3dB CTLE?

General EIC questions

- We will use LM stabilization IP
 - What is the best way? Hard maco?
- What are the ucontroller specs?
- UCle mux gearbox
 - Does it support back pressure? What assumptions are being made?
- Will this support both OCigen1 and gen2?

DO NOT SHARE

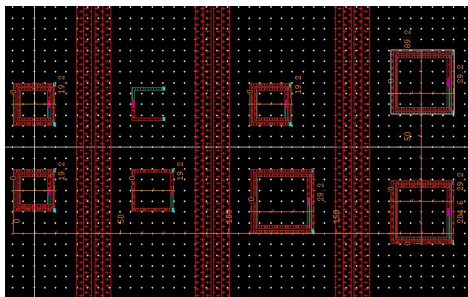
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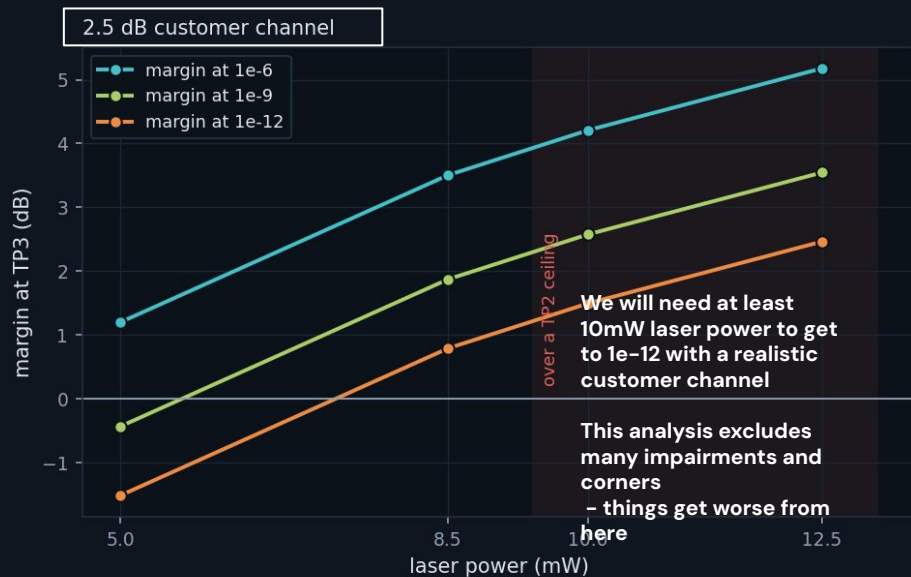
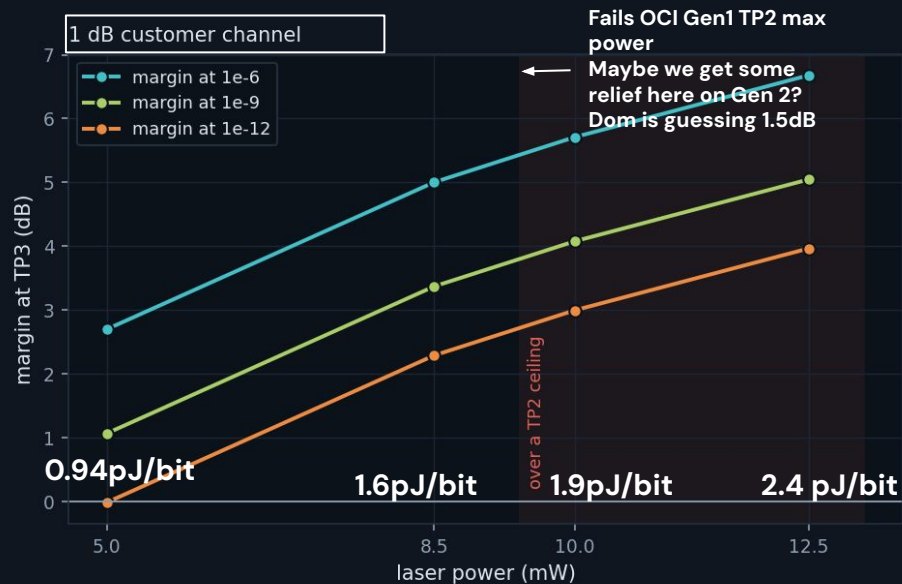


Gizmo TIA
floorplan from Cliff
Ting

- 270umx540um = 0.146mm² for 4 lanes including DRV/TIA
 - **0.0365mm²/lane**
- **Comparisons**
 - Gizmo target < 0.15mm²/lane excluding DRV/TIA
 - 224G SerDes: 0.6–0.7mm²/lane
 - Gizmo TIA: 205x90 = 0.018mm²
 - Multi stage cherry hooper with inductors
 - Gizmo DRV: 0.04mm²
- One entire lane from Quebec is smaller than our driver
- **Highly unlikely that this is actually realizable – more realistically this is a bump limited floorplan**

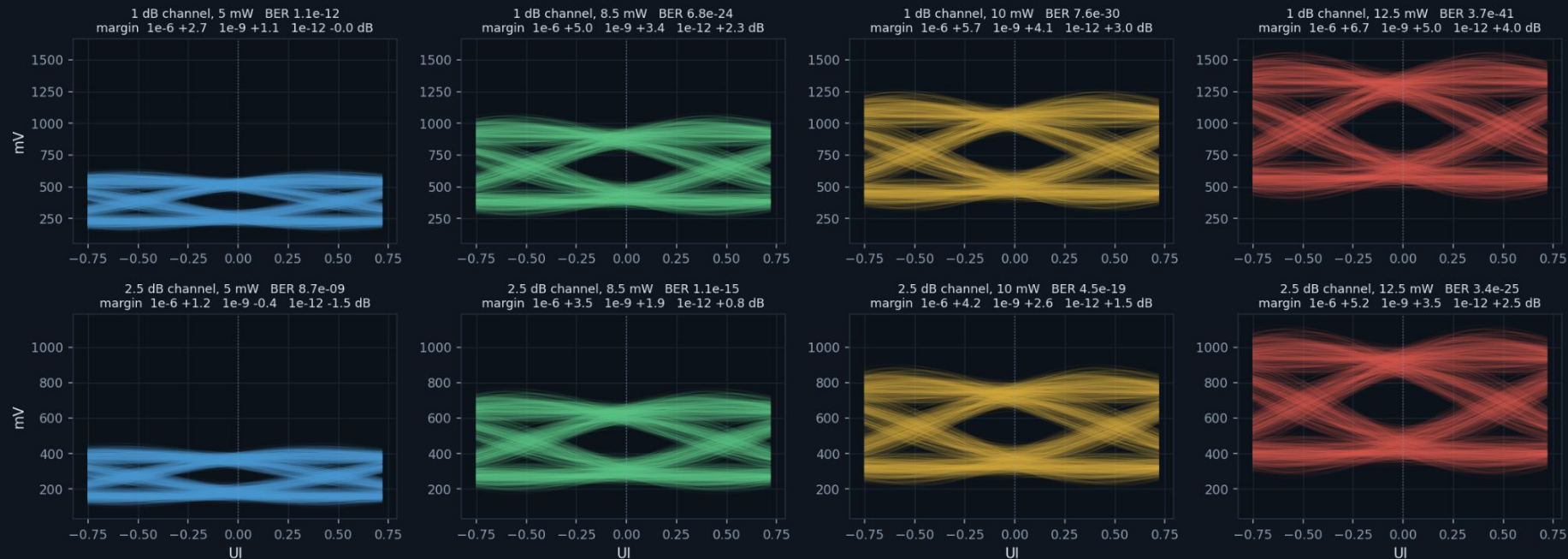
Gizmo: BER Performance

Link margin vs laser power, simulated: 106.25 GBd, ring 150 pm, 2.8 Vpp pre -0.12, case_1, 4.25 + channel + 4.25 dB, no RX equalisation



Eyes

Slicer eye per cell with measured TIA noise and shot noise (8,192 symbols)



Transmitter

Transmitter electrical specs

- Pad, ESD, and routing parasitic must be accounted for.
- Closely placed driver & Tx optics
- Needs further discussion

Parameter	Value	Units
Driver electrical 3-dB BW	65	GHz
Driver swing	2.5	Vppd
Reverse bias	1.25	V
TX FFE	2	Taps

Targets are quite similar to Gizmo –
2.5V–3V swing effective, 3 Tap FFE
(main + 1 pre– + 1 post)

Receiver

Receiver electrical specifications

- TIA performance is highly sensitive to input capacitance.
- Recommend next step to define exact EIC pad/ESD capacitance and electrical routing to TIA

Parameter	Value	Units
Data rate	100	Gb/s NRZ
BER target	1E-12	—
Photodiode reverse bias	2 ~ 3	V
Responsivity	≥0.85	A/W
Photodiode capacitance	≤20	fF
Electrical Receiver Small Signal Bandwidth (inc. PD)	50	GHz
Target input RMS noise	3.1-4	μA rms
CTLE	3	dB at Nyquist
CTLE programmability	1	dB steps
Input-referred noise density	15-20	pA/√Hz
OMA sensitivity target (at PD plane)	-9.5	dBm OMA

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Spec row	Quebec value	Gizmo Cliff Ting's TIA
Data rate	100 Gb/s NRZ	106.25 Gb/s NRZ
BER target	1e-12	1e-9 is the likely LZ
Responsivity	≥ 0.85 A/W	0.9 assumed
PD capacitance	≤ 20 fF	TBD
Small-signal bandwidth incl. PD	50 GHz	50.2 GHz, no PD
Input RMS noise	3.1 to 4 μA	3.5 μA full band, 2.7 to 3.1 μA to f_3dB
CTLE	3 dB at Nyquist, 1 dB steps	0.33 dB of fixed peaking, no CTLE
Input-referred noise density	15 to 20 pA/√Hz	8 at 1 GHz, 16 at 25 GHz, 19 at 50 GHz
OMA sensitivity at PD, 1e-12	-9.5 dBm	~ -7.5 dBm Unlikely that we will actually land at 1e-12 with impairments

Gizmo: Data-Path Insight Enables Better MRM Control

- **Pattern-dependent self-heating is the MRM control problem at 106.25 GBd**
 - Deterministic, zero-mean resonance wander driven by run length / disparity; thermal response extends from tens of ns to $\sim\mu\text{s}$
 - Budget is tight: OCI Gen1 dTDEC (SSPR vs. PRBS13) ≤ 0.4 dB; 400G mode absorbs 3 dB more power on the same allocation
- **Today's observables cannot see it**
 - L2V average-power lock: DC-balanced data has zero mean shift—reports "locked" while the eye closes; cannot separate a disparity change from resonance drift
 - PGT dither lock-in: bandwidth is bounded by dither / integration (kHz class), while dither amplitude spends TDEC budget
 - Both are feedback through the thermal plant and observe the drop port after heat moves
- **Gizmo owns the serializer—the pattern is known before it reaches the ring**
 - TX disparity checker is already in the Gizmo architecture, with its heater-servo gating defined; the feedforward path is the next step
 - Feedforward: filtered running disparity $\rightarrow$ pre-emphasized heater code + bias-VDAC term for the ns edge; drop-port power becomes calibration, not the only error signal
 - Precedent: Sun et al., JSSC 2016—bit-statistics tracking + self-heating cancellation held the eye open under arbitrary disparity
- **AthenaCore hides the serializer; off-the-shelf requires a TX disparity/data-path tap**
 - Export running disparity to the photonic controller—or give up this control advantage

Gizmo: Squelch Is a Known Event, Not a Disturbance

- **Squelch is a mandated MRM plant change on every relink (OCI Gen1 Table 1-3)**
 - RF removed, OMA ≤ -15 dBm (200G) / -12 dBm (400G), average power held within ± 0.5 dB, 60-75 ms dwell, then straight into the 160-bit deskew training pattern
 - Removing modulation changes intracavity energy and carrier self-heating -> a resonance step the heater must absorb with zero unlock events (≤ 10 ms target)
- **To a drop-port-only controller (L2V / PGT) squelch looks like sudden drift**
 - Servo chases the step; dither leaks into the squelched-OMA margin; a wrong-direction excursion risks a spurious far-end LOS
 - Exit lands on the training pattern while the ring is still walking back—pattern heating resumes on top of the step
- **Gizmo's squelch sequencer commands the event: instant, direction, V_sq, exit target**
 - Already architected: heater ramp-limit freeze at entry/exit, disparity path gated so the static HOLD code is not read as data, V_sq calibrated per lane against the drop port
 - Enables a mode switch: feed forward the calibrated heater offset at entry/exit, run average-power lock only while the pattern is static, return to disparity feedforward at exit
 - Constant-AOP squelch needs independent swing-mute and DC-bias paths in the driver—a Gizmo driver decision, not a generic SerDes electrical idle
- **AthenaCore executes squelch inside the macro—we would see only the optical aftermath**
 - Ask: export squelch state / timing / exit target; confirm bias-hold squelch (not swing collapse or laser blanking) with swing-independent DC bias

Bump Map

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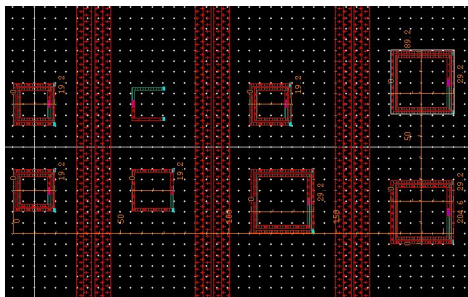
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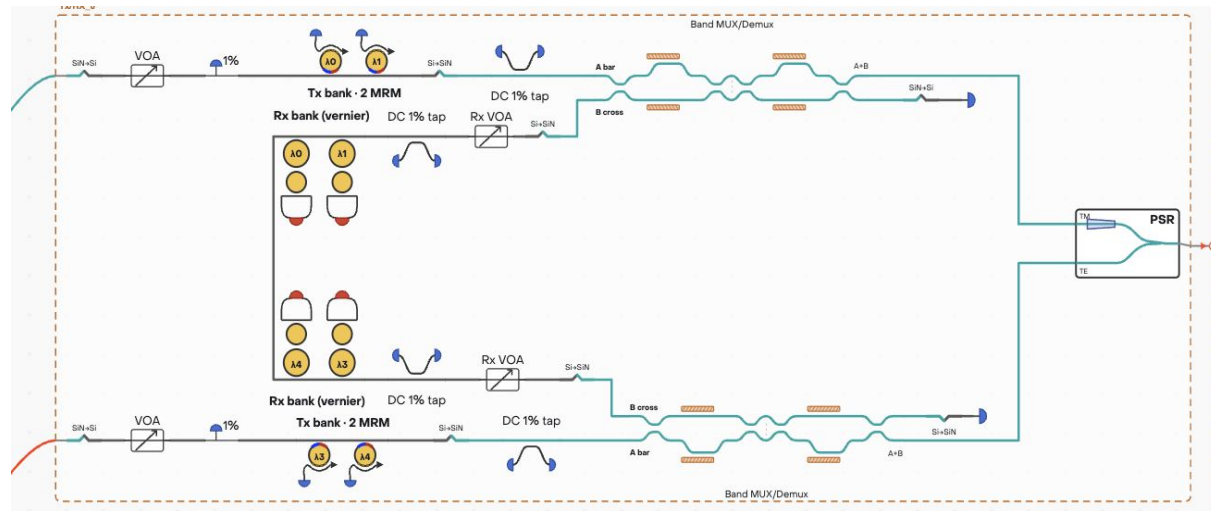
- **COUPE OCI-GEN2**
 - LM/MA Lane Pitch = 450um
 - QB => 135um
- **There is no way to meet such a tight pitch**
- **Bank Needs Depopulated bumps to fit CRRs and associated path length matching**
- **TSV/C4 Bump pitch and Hybrid bump pads needs to be relatively sparse**
- **Waveguide Density is a concern**

This does not include additional space required for analog KOZs



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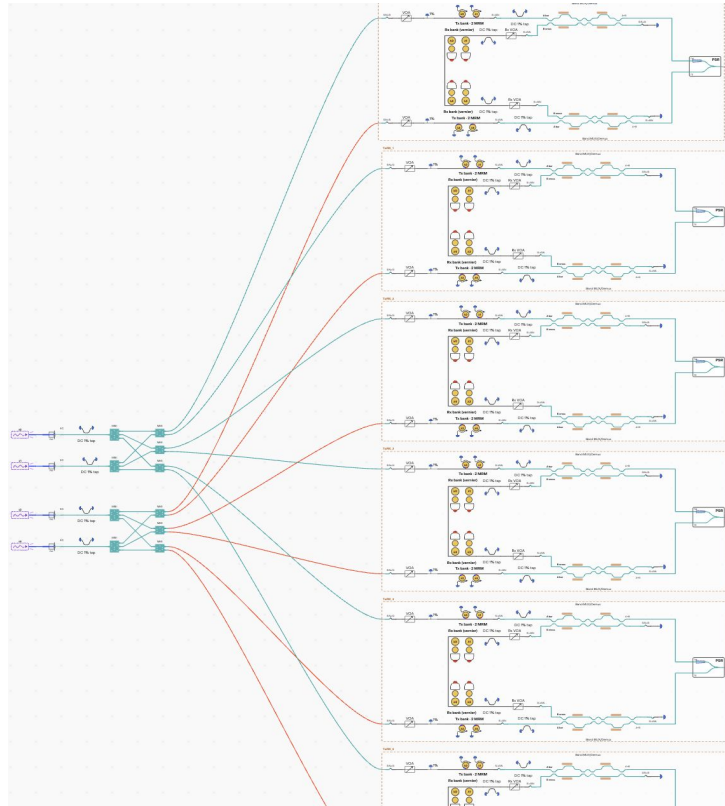
For comparison:



Pn dither	4	Per bank
mPDs	16	Per Bank
INTL Heater	8	Per bank
MRM heater	4	Per bank
CRR heater	8	Per bank
Pol Cntrl Heater	6	Per bank
mPD per laser	1	Per bank

Assume architecture mitigating NLA and 4-wave mixing and mitigates backreflections through an interleaver on both sides

As well as high power MRMs



AFE section	Pad type	Count
TX	TX output, high speed	2
TX	DAC output	1
TX	MPD signal	1
TX	MPD Supply	1
TX	Spare	1
TX	Driver supply (VDDH)	1
RX	RX inputs, high speed	2
RX	DAC outputs	2
RX	PD Supply	1
Common	Grounds	4
Common	VDDL	1

Note: 6 additional Grounds at Tile level

Under counting the number of bumps